PARKINSON'S PROGRESSION MARKERS INITIATIVE

Play a Part in Parkinson's Research

Proteomics analysis of peripheral blood mononuclear cells

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Summary

1203 peripheral blood monuclear cell (PBMC) samples were received from PPMI and analysed by mass spectrometry. In short, the cells were isolated by centrifugation from frozen ampules and analyzed with liquid chromatography tandem mass spectrometry (LC-MS/MS) in data independent acquisition (DIA) mode, to study Parkinson's disease (PD) biomarkers and progression markers. This version of the method document includes steps performed up to LC-MS/MS analysis.

Method

Protein sample preparation for LC-MS/MS analysis

Samples identifiers were blinded, and samples provided in cryopreservation medium. Cryopreserved samples were thawed at 37°C in water bath for 3 minutes. Supernatant was removed by centrifugation, and thawed samples were lysed in SDS containing lysis buffer with 300U/ml Benzonase and prepared according to STrap™ manufacturer's recommendations (https://protifi.com/pages/s-trap). Following lysis and loss of viscosity from breakdown of DNA by Benzonase, the samples were incubated 5 minutes at 95°C. Following heating 3 µl of each sample was collected for protein concentration determination. Each sample preparation batch contained a control sample, which is from a single buffy coat isolation, which was aliquoted as cell pellet and stored at -80°C. Following lysis and protein concentration determination sample collection, samples were frozen on dry ice and stored at -80°C until sample digestion.
Protein concentration was normalised and samplesamples with internal code A-K were digested on 96-well S-Trap™ (ProtiFi, LLC, Fairport, NY) plates with Microlab star (Hamilton Company, Reno, NV) robot aid. For samples L-N and redigestion samples, the samples were prepared manually using S-Trap™ micro spin columns (ProtiFi), because the protein amount was limited. 100 µg of protein was used for trypsin digestion in 96-well plate format. Up to two 96-well plates were digested on the same day, and these batches were A&B, C&D, E&F, G&H, I&J. Each 96-well plate included two buffy coat samples, that were lysed on the same day as the digestion batch samples (as controls), and are referred as digestion standards.
Digestion followed S-TRAP manufacturer's recommendations (https://protifi.com/pages/s-trap). Following mixing, 90% methanol solution containing 100 mM Tris-HCl pH 7.1 (loading buffer) was added to samples, and samples were loaded on to 96-well plates. After loading samples were washed four times with loading buffer. Sequencing grade Trypsin (Promega Corporation, Madison, WI) was added to samples: 2 µg for 96-well plate samples and 1 µg for micro column

samples in 50 mM Ammonium bicarbonate. Following trypsin addition, samples were incubated for 18 h at 37℃ in humified incubator. Samples were eluted according manufactures instructions and vacuum died to total dryness with Speedvac (Thermo Fisher Scientific, Inc., Waltham, MA) and stored at -80℃ until analysis.

LC-MS/MS analysis

Samples in 0.1% formic acid were normalised for peptide concentration and run according to Evosep manufacturer recommendations using the 30 sample per day gradient. Following peptide yield determination 800 ng of each sample was spiked with index Retention time peptide pool (Biognosys AG, Schlieren, Switzerland) and loaded on EvoTip (Evosep Biosystems, Odense, Denmark) according to manufacturer's instructions. Daily run standard control, which is a pooled buffy coat sample prepared from a 96-well S-trap digestion was prepared in same way. Evosep One (Eveosep Biosystems) was operated in 30 samples per day mode and interfaced with Obitrap Lumos mass-spectrometer with FAIMS-pro (Thermo Fisher Scientific). Orbitrap Lumos was set to acquire MS1 scan at 120 000 resolution, and MS2 scans at 30 000 resolution, and FAIMS device was operating at – 40 compensation voltage (CV). MS2 scans were acquired in DIA mode with 32 variable width windows with normalized AGC target of 800% and maximum 54 ms ion fill time.

QC and sample re-runs

Each digestion with a unique internal standard label had two buffy coat samples and are referred as A-L/redigestionCtrl1/2. Additionally daily at least one buffy coat run standard referred as "date_DF_buffycoat_x", where x denotes number of technical injection. These samples are used to asses daily performance of the MS. Cohort sample labelling nomenclature follows: "date_DF_internal sample code_1".

In total 1203 samples were received from PPMI. Applied internal labeling annotation can be found in the following file: "160523_DF_PPMI_internal_codes.xlsx". Samples that did not pass QC during the MS run and re-run are labelled "sample not successfully analysed" in notes column.

Specifically the following samples (internal codes) had quality issues due to unstable electrospray were re-run: A56, A57, B48, C75, D67, D73, E42, E43, E44, E53, F1, F2, H8, H9, H11, H16, H23, H24, H44, H53, H54, H70, H72, H73, H74, H77, H78, H79, H80, H81, H82, H85, H86, H87, I83, I86, J43, J85, L74. Total of 39.

Samples with following internal codes had failed digestion and were used to design a redigest batch with S-Trap microcolumn followed by MS re-run: B15, K61, K62, K63, K64, K65, K66, K67, K68, K70, K71, K72. Total of 12.

In total, out of 1203 samples analysed, 5 did not pass final QC even after re-digestion/re-run. This information is found on the excel file.

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Data Access

To access data, please download and complete the 'PPMI Large Datasets Request (Non-Genetic) Form' located in the 'Study Data' section of PPMI IDA. Submit your request, along with the completed form, to ppmi@loni.usc.edu with the subject line "PPMI Large Datasets Request (Non-Genetic)".

About the Authors

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